Video Imaging

Dr. Nelson Díaz
Departamento de Ingeniería Informática
Universidad Técnica Federico Santa María, campus Valparaíso, Chile

nelson.diazd@usm.cl

Academic website https://nelson-diaz.com/

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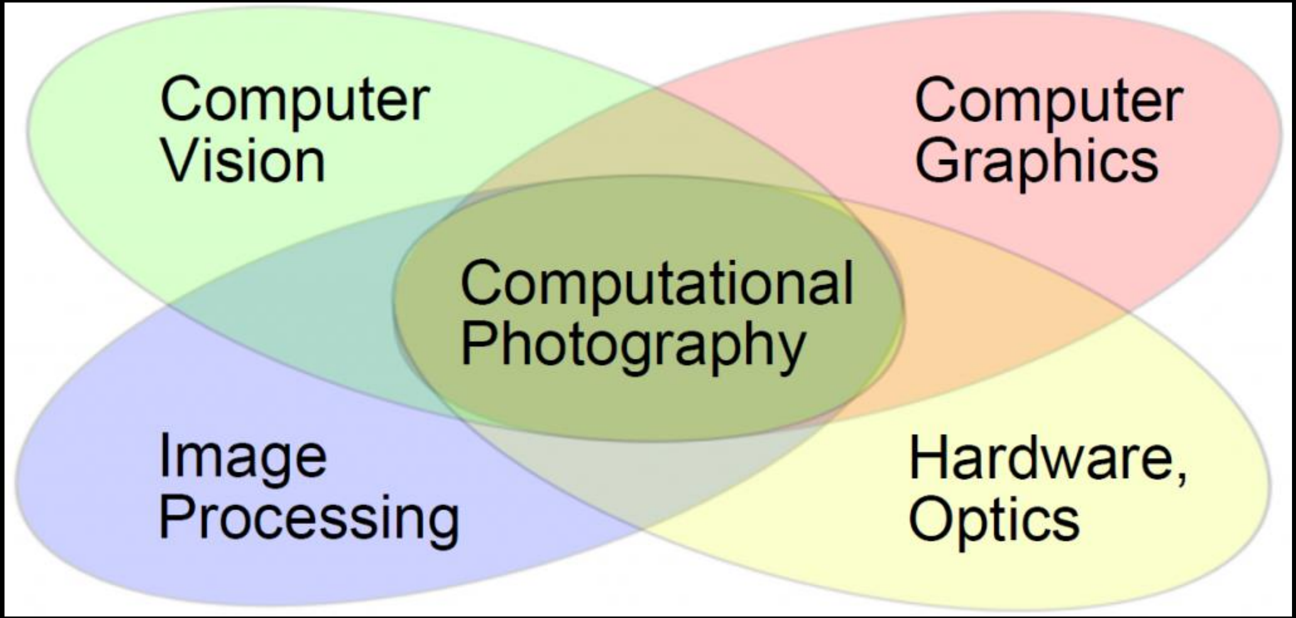
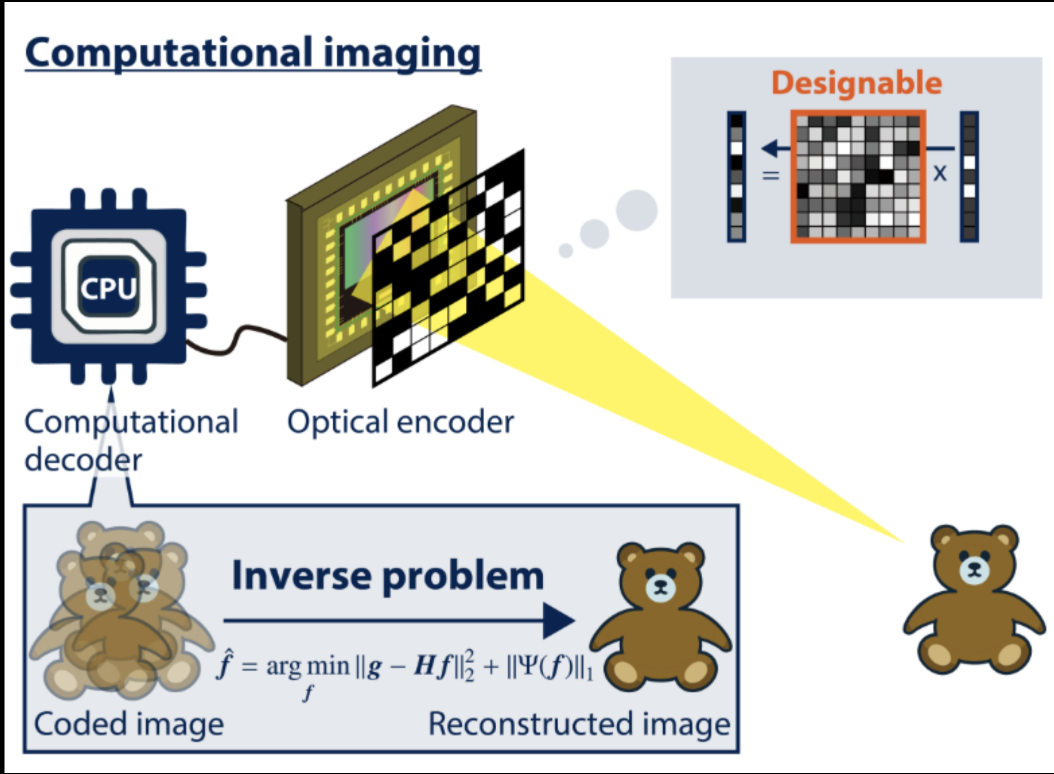
Computational Photography



Areas:

- Computer Science
- Electrical Engineering and Electronics
- Maths
- Physics (optics)

Computational Photography



Goal:

- To capture and recover singal in many dimensions

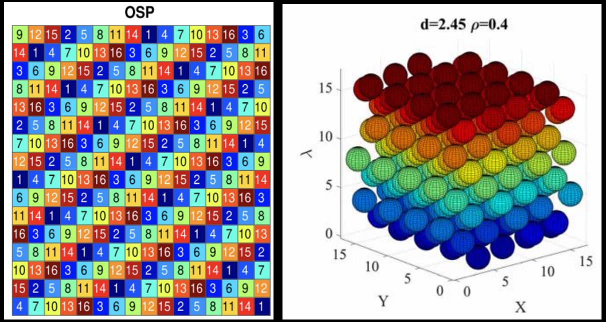
To design

- Hardward
- Software--algorithm to solve the inverse problem--

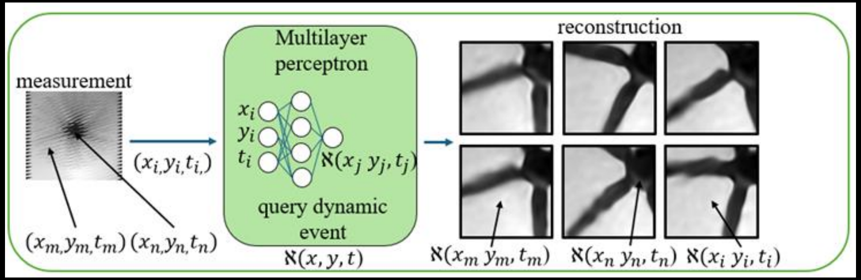
Computational Photography: Applications

- It can be applied to different signals
- It use compressive measurements
- To solve an inverse problem

Multispectral

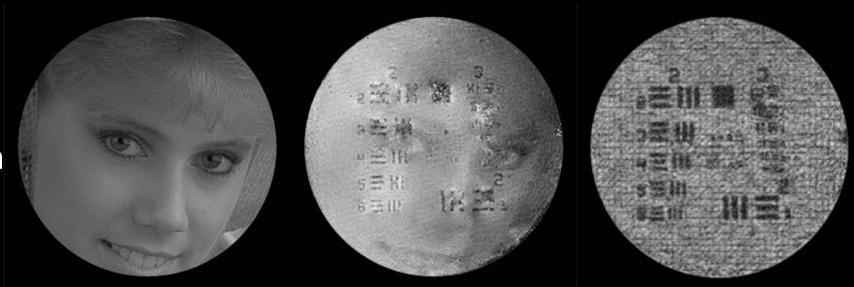


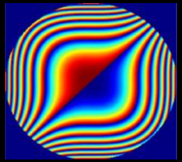
Video



GT phase | Recovered phase | Recovered Amplitude

A woman





1995

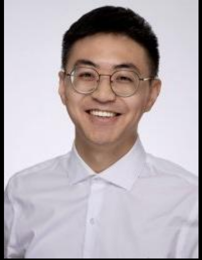
Cubic Phase

Depth

Single-mask sphere-packing with implicit neural representation for ultrahigh-speed imaging







Nelson Diaz[1], Madhu Beniwal[2], Miguel Marquez[2], Felipe Guzman[1], Cheng Jiang[2], Esteban Vera[1], Jinyang Liang[2]

[1] School of Electrical Engineering, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile
[2] Laboratory of Applied Computational Imaging, Centre Énergie Matériaux Télécommunications, Institut National de la Research Scientifique, Montreal, Canada

INRS Institut national de la recherche scientifique

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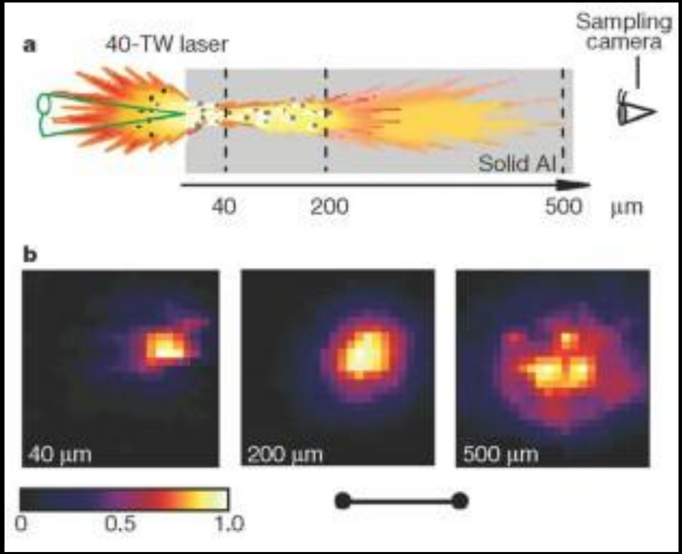
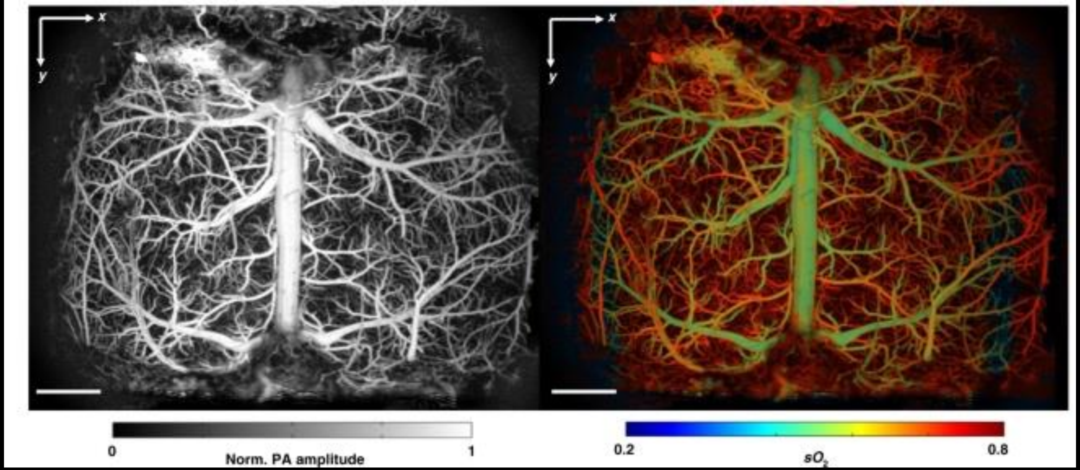
OPTOLAB

Sensing Transient Events



Nuclear fusion: Fast-heating plasma



Neuroscience for monitoring neural activity

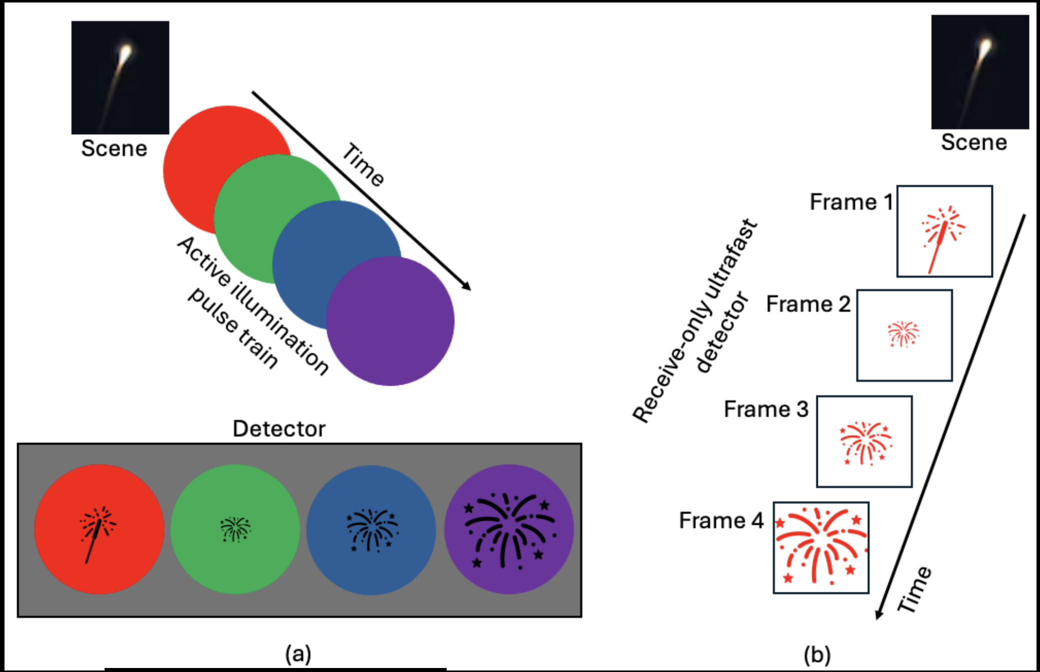
Conventional cameras capture up to 10^7

R. Kodama, P. Norreys, K. Mima, A. Dangor, R. Evans, H. Fujita, Y. Kitagawa, K. Krushelnick, T. Miyakoshi, N. Miyanaga et al., "Fast heating of ultrahigh-density plasma as a step towards laser fusion ignition," Nature 412, 798–802 (2001)

X. Zhu, Q. Huang, A. DiSpirito, T. Vu, Q. Rong, X. Peng, H. Sheng, X. Shen, Q. Zhou, L. Jiang et al., "Real-time whole-brain imaging of hemodynamics and oxygenation at micro-vessel resolution with ultrafast wide-field photoacoustic microscopy," Light. Sci. & Appl. 11, 138 (2022).

Active and Passive illumination

- A pulse probe for a transient event.
- An optical marker e.g. spatial division, or angle division divide the transmitted probe pulses
- After the transient scene in the spatial frequency domain to recover the 3D spatio-temporal datacube.



Active-detection

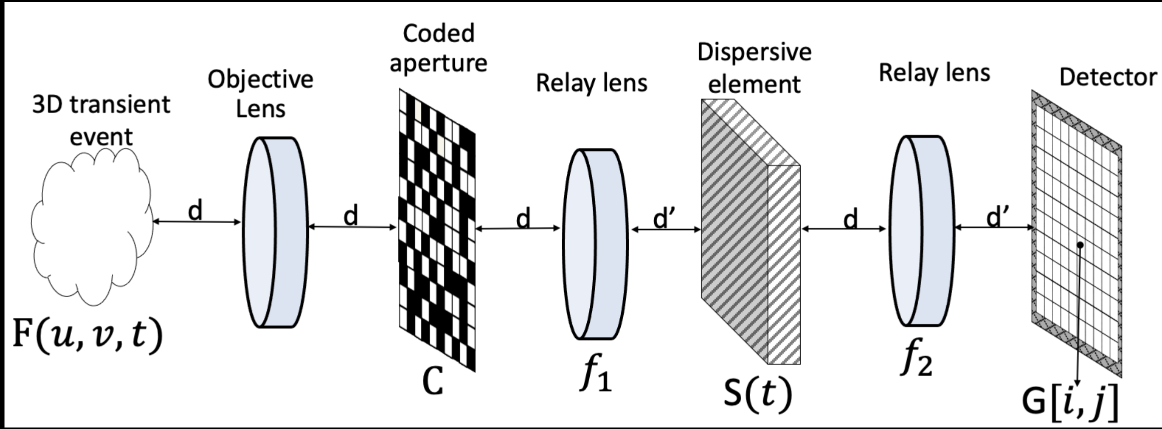
Passive-detection

- Passive detection can overcome active illumination.
- The detector only receive and record the emitted and/or scattered photons from the scene.

Jinyang Liang and Lihong V. Wang, "Single-shot ultrafast optical imaging," Optica 5, 1113-1127 (2018)

Single snapshot compressive spectral imaging

The scene is encoded using a binary coded aperture. Then, the encoded scene is sheared by the temporal shearing operator. The encoded and sheared scene is captured in the detector.

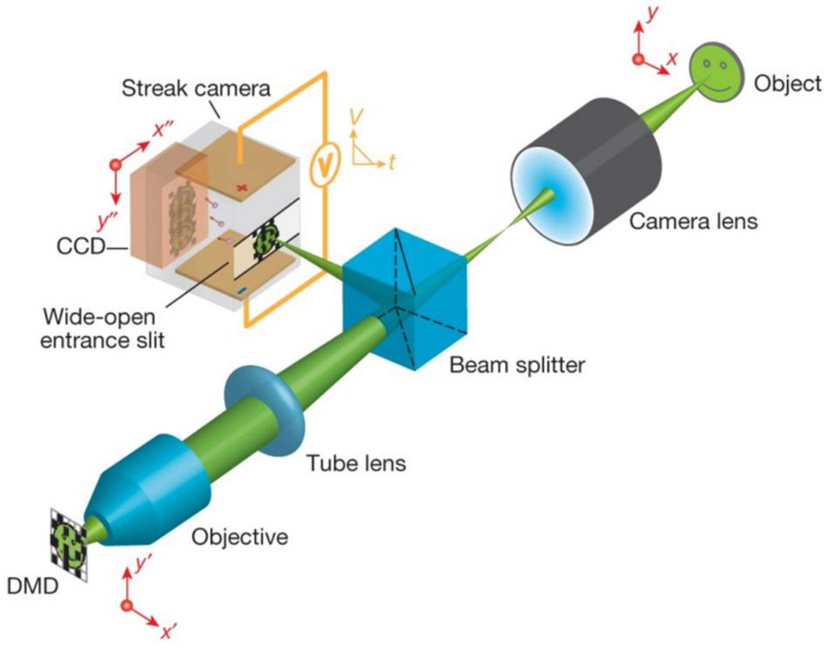


$$\hat{f} = \Psi^{-1}\left(\underset{\theta}{\operatorname{argmin}} \frac{1}{2}\|y - H\Psi\theta\|_2^2 + \tau\|\theta\|_1\right)$$

Ashwin Wagadarikar, Renu John, Rebecca Willett, and David Brady, "Single disperser design for coded aperture snapshot spectral imaging," Appl. Opt. 47, B44-B51 (2008)

Compressive Ultrafast Photography (CUP)

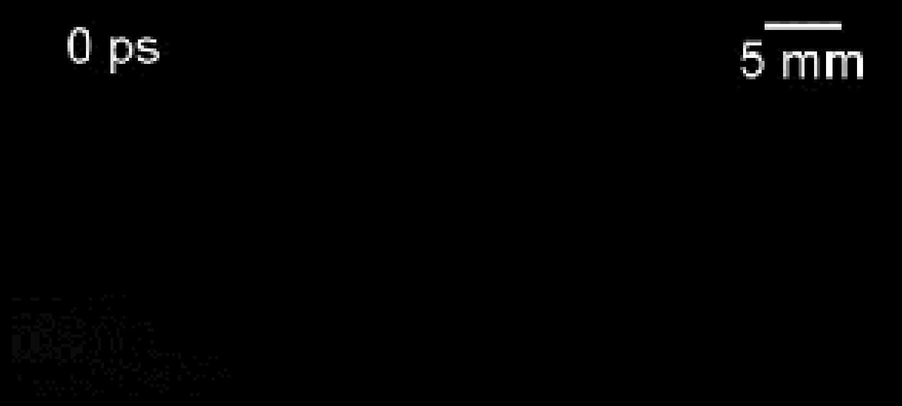
The scene is encoded using a binary coded aperture. Then, the encoded scene is sheared by the temporal shearing operator. The encoded and sheared scene is captured in the detector.



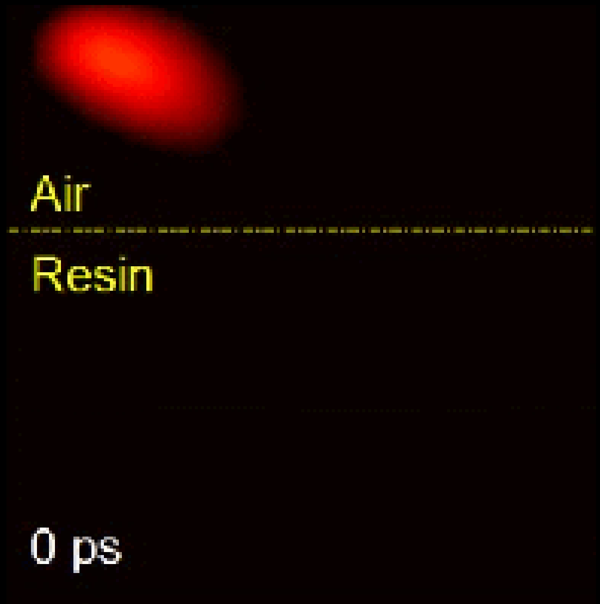
- It is a passive-detector approach

L. Gao, J. Liang, C. Li, and L. V. Wang, “Single-shot compressed ultrafast photography at one hundred billion frames per second,” Nature 516, 74–77 (2014).

Compressive Ultrafast Photography (CUP) Applications



Optical Mach cone using CUP

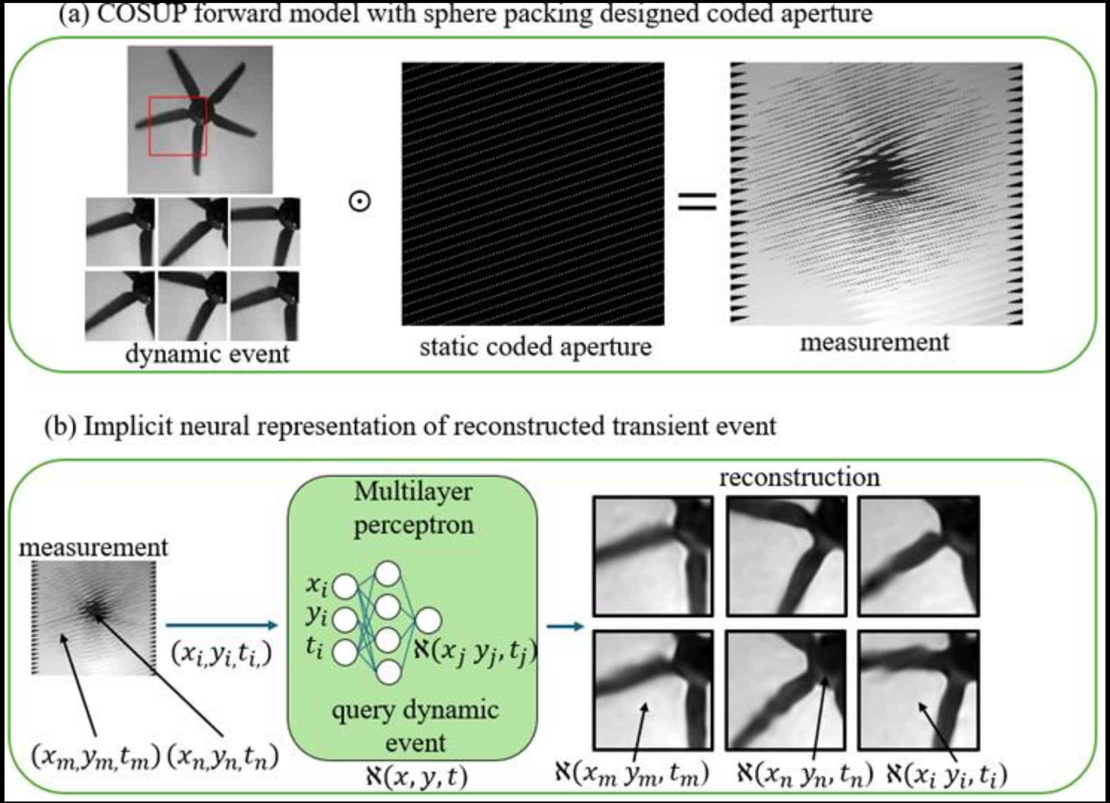


Single-shot CUP pulse refraction

Real-time applications is limited by reconstruction

L. Gao, J. Liang, C. Li, and L. V. Wang, “Single-shot compressed ultrafast photography at one hundred billion frames per second,” Nature 516, 74–77 (2014).

Representación neuronal implicita para video ultrarápido

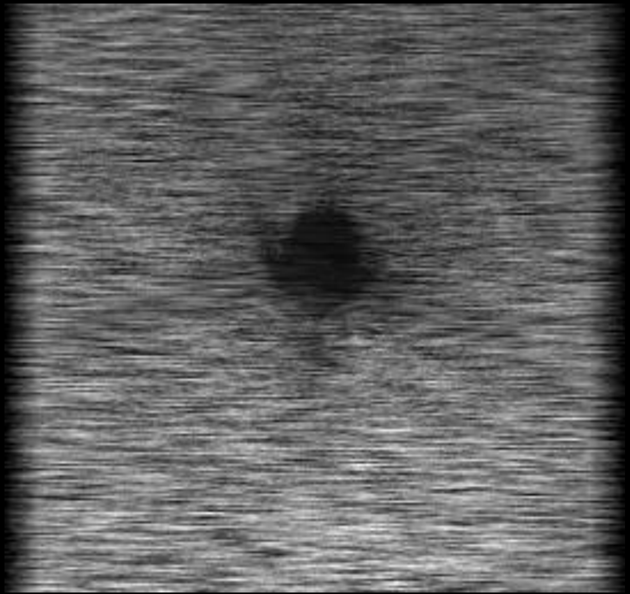


N. Diaz, M. Beniwal, M. Marquez, F. Guzmán, C. Jiang, J. Liang, and E. Vera, "Single-mask sphere-packing with implicit neural representation reconstruction for ultrahigh-speed imaging," *Optics Express 33,* pp. 24027–24038, 2025.

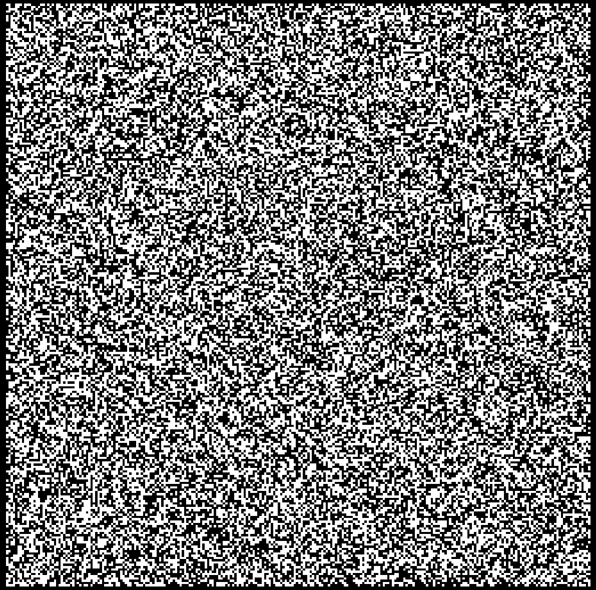
COSUP Discrete Model

$$Y_{(:,\bar{t})} = \sum_{t=0}^{T-1} X_{(:,:,\bar{t})} \odot C_{(:,:,\bar{t})} + \Omega$$

designing the encoding is critical
for real-time applications



measurement



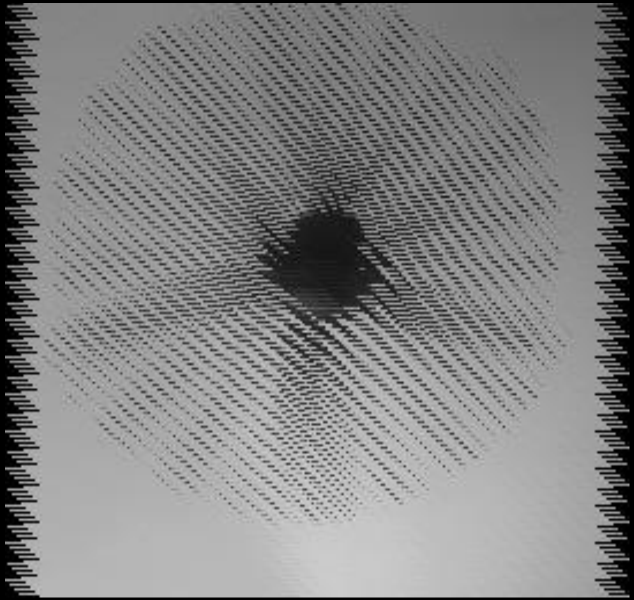
traditional coded aperture

video

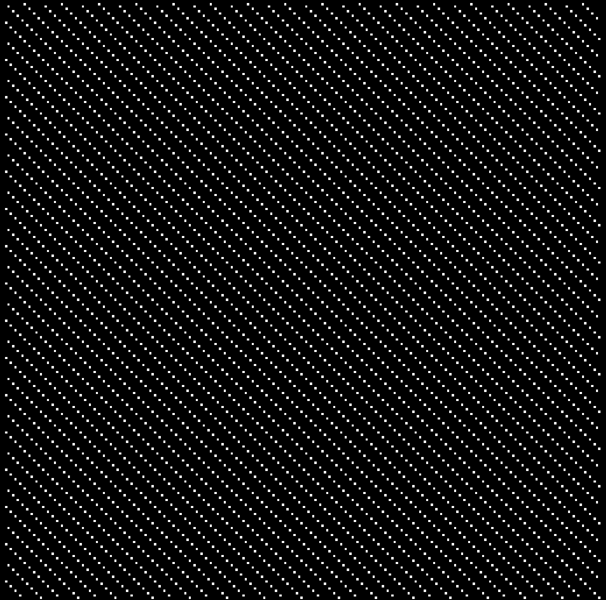
Discrete model

$$Y_{(:,\bar{t})} = \sum_{t=0}^{T-1} X_{(:,:,\bar{t})} \odot C_{(:,:,\bar{t})} + \Omega$$

novel aperture design



measurements



coded aperture

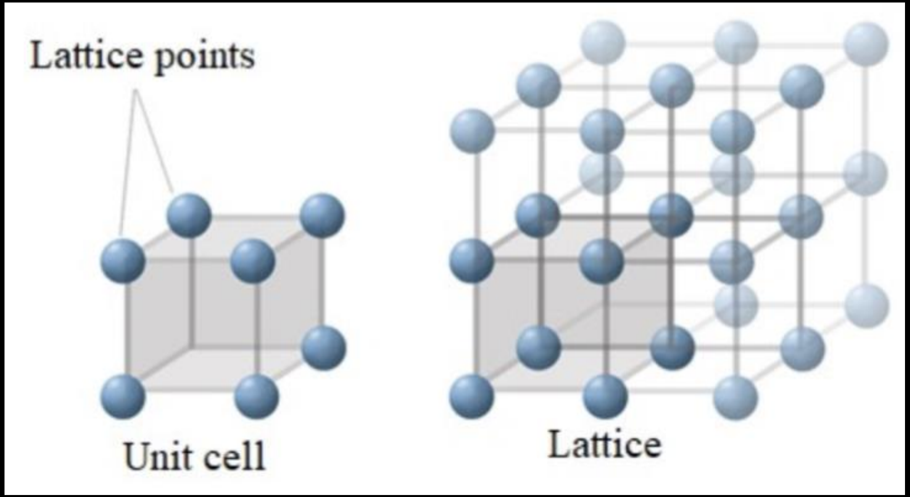
video

What is Sphere Packing?

- Ask for the densest packing of R^n
- Faced-center cubic lattice is optimal in 3D

$$\rho_{\Lambda_3} = \frac{\pi}{\sqrt{18}} \approx 0.74 \ldots$$

stacking cannonballs

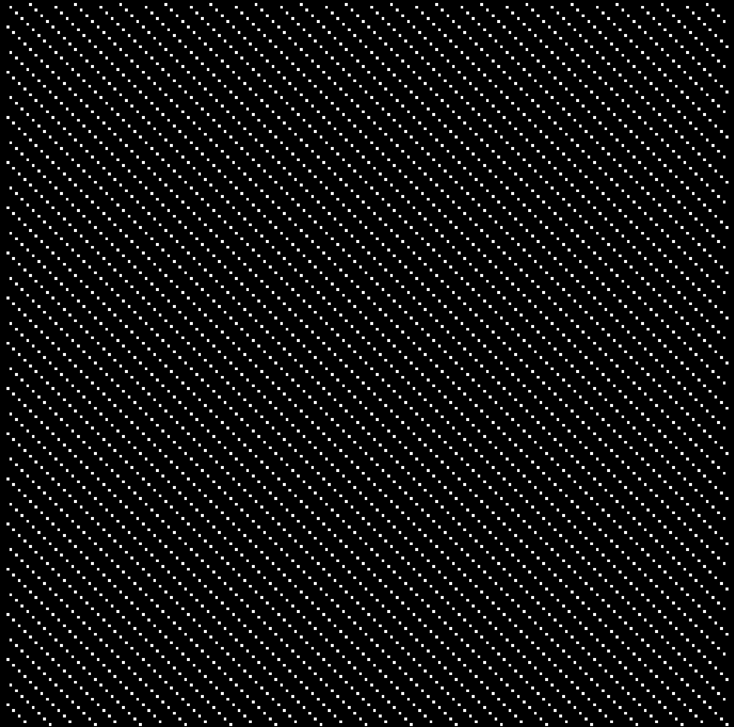


Tiling the space by replicating the unit cell

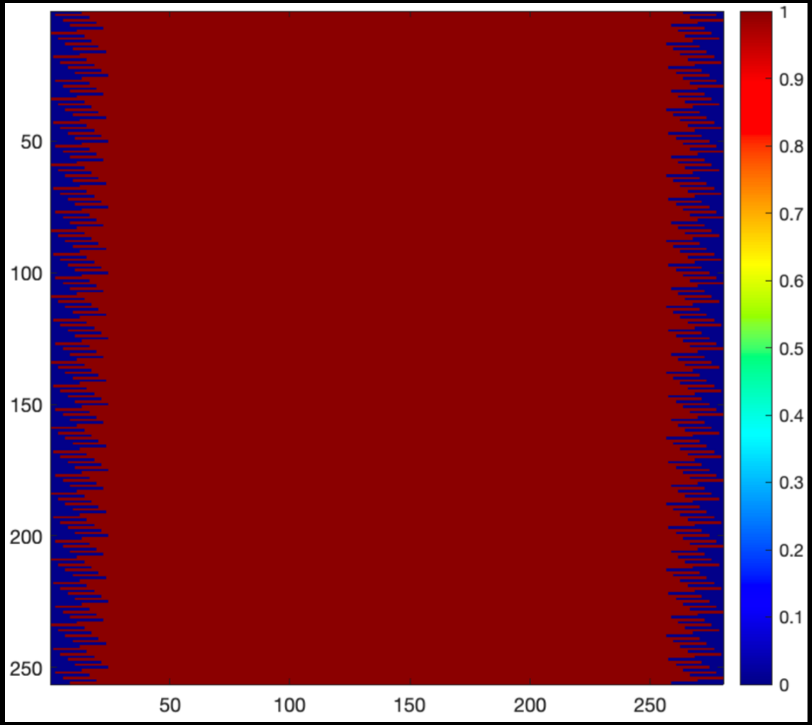
J. Kepler, The six-cornered snowflake (Paul Dry Books, 2010).
T. Hales, M. Adams, G. Bauer, T. D. Dang, J. Harrison, H. Le Truong, C. Kaliszyk, V. Magron, S. McLaughlin, T. T. Nguyen, and et al., “A formal proof of the kepler conjecture,” Forum Math. Pi 5, e2 (2017).

Coded Aperture Design

- The coded aperture exploits the shifting induced by the galvanometer scanner.
- Our design guarantees uniform sampling.

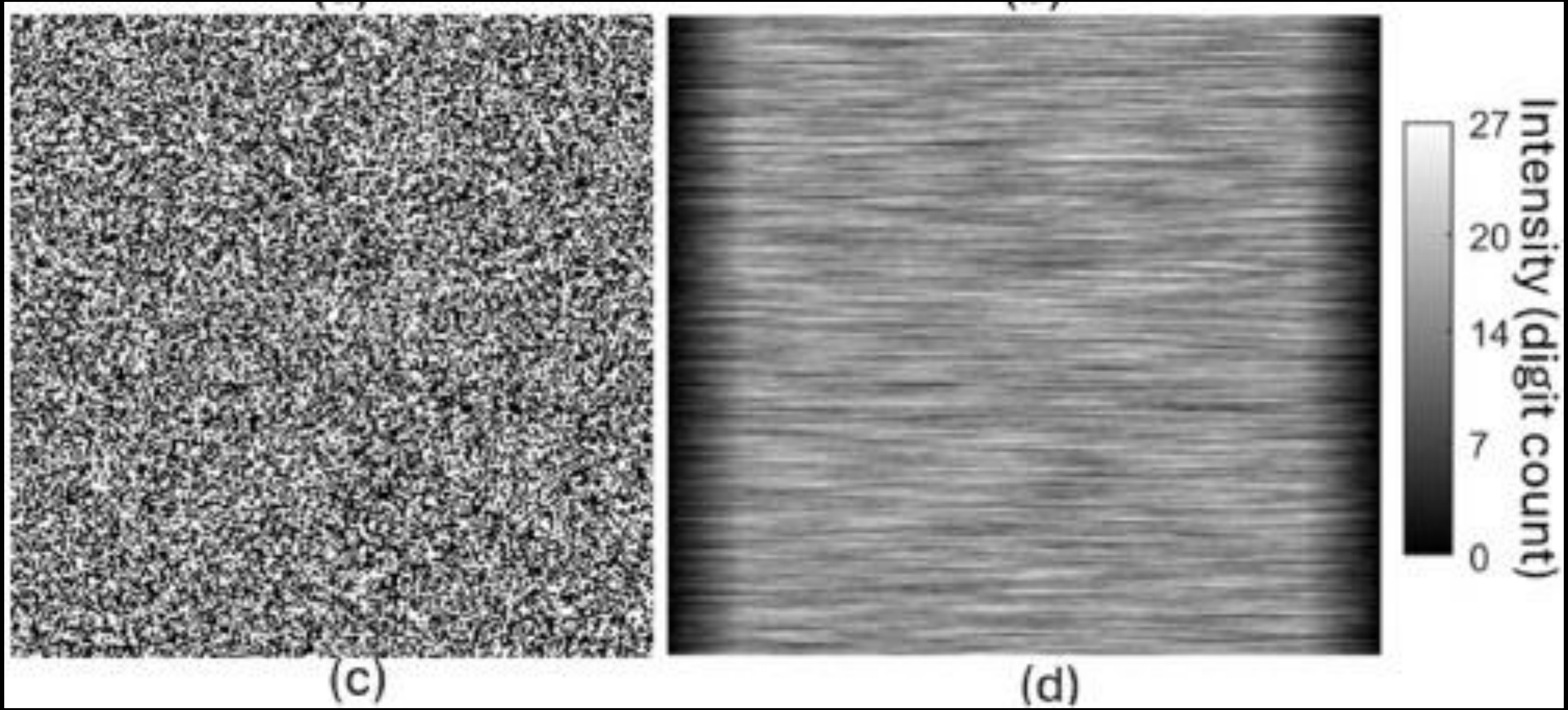


uniform coded aperture



uniform sensing

Traditional Coded Aperture Design



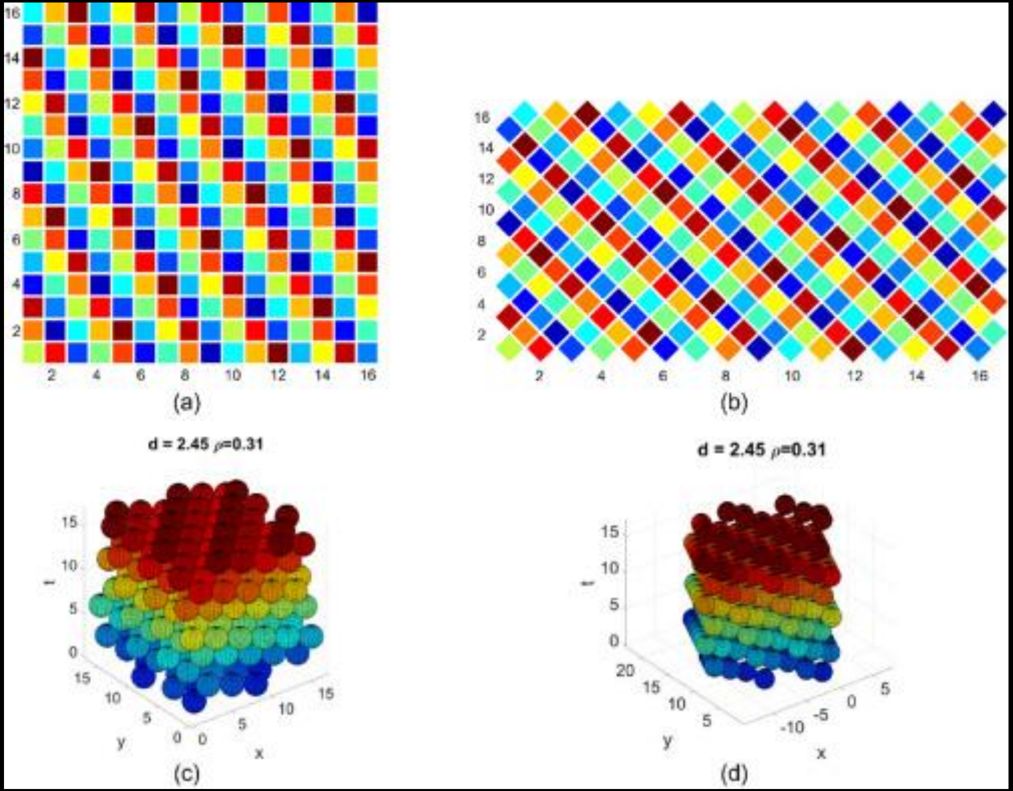
Random coded aperture

Non-uniform sensing

Sphere Packing Comparison Between DMDs

orthogonal DMD

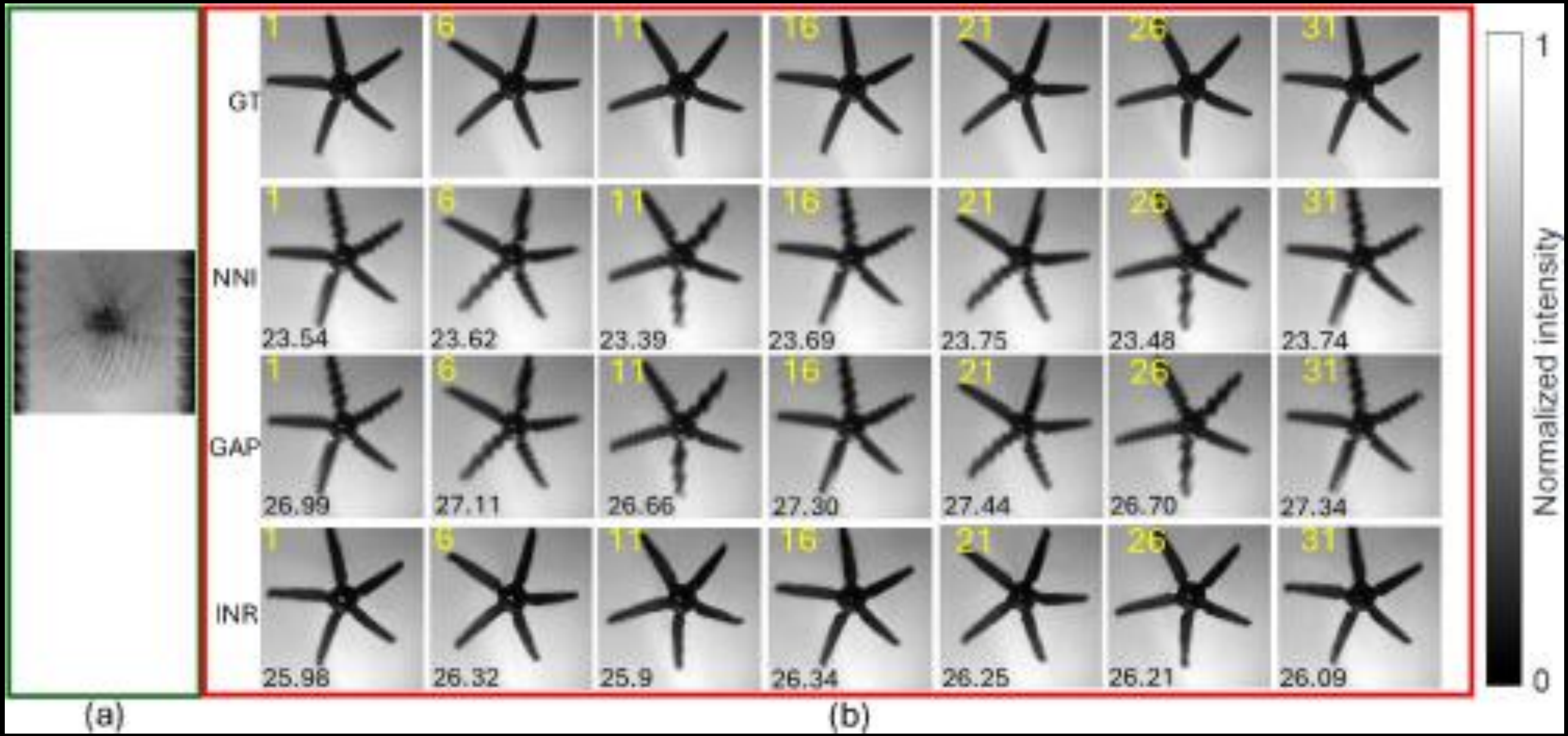
diamond DMD



Texas instruments

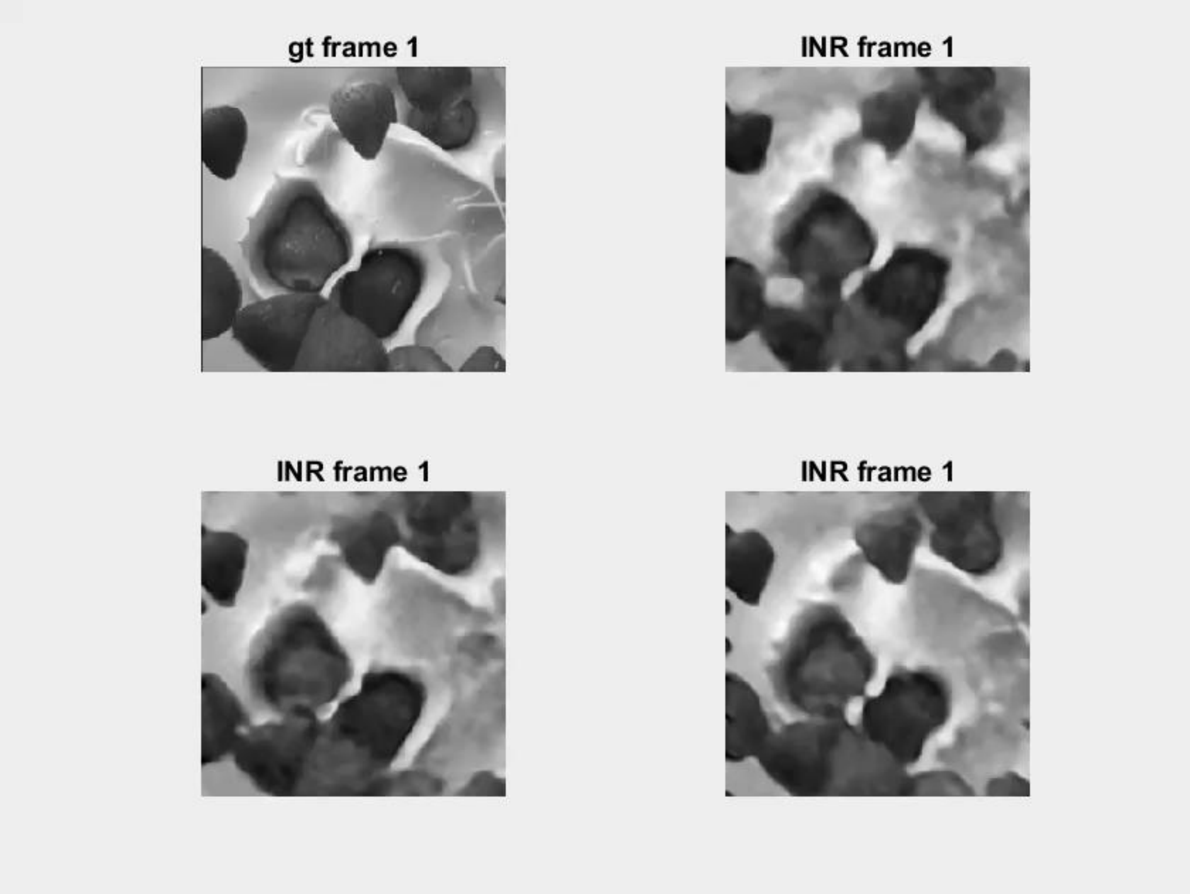
orthogonal: DLP3010LC
diamond: DLP® 0.45 WXGA DMD

Simulation Results



Simulation Results

Simulation Results



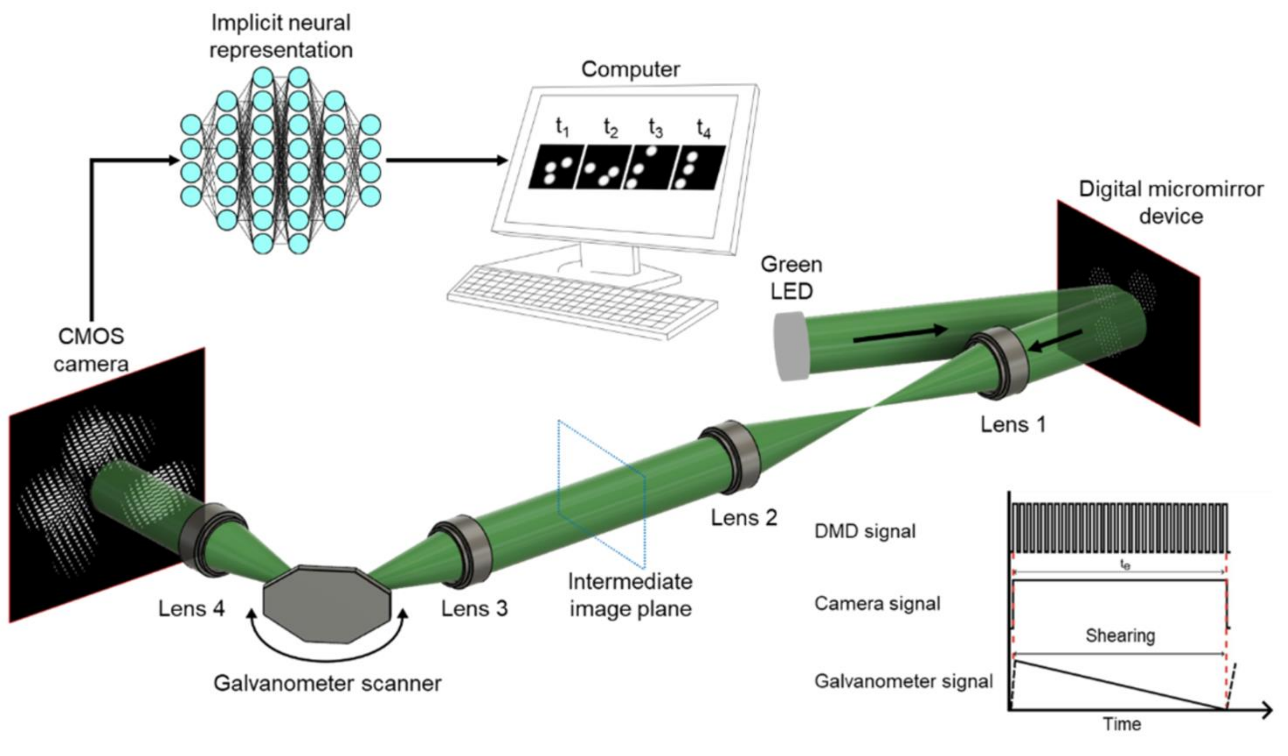
Simulation Results

measurements reconstruction measurements reconstruction

Algorithm	CA	Number of frames		
		$T = 16$	$T = 25$	$T = 31$
NNI	random CA PSNR (dB)	27.59	25.75	24.98
	random CA SSIM	0.91	0.87	0.83
	SPCA PSNR (dB)	28.33	26.9	25.06
	SPCA SSIM (dB)	0.92	0.89	0.85

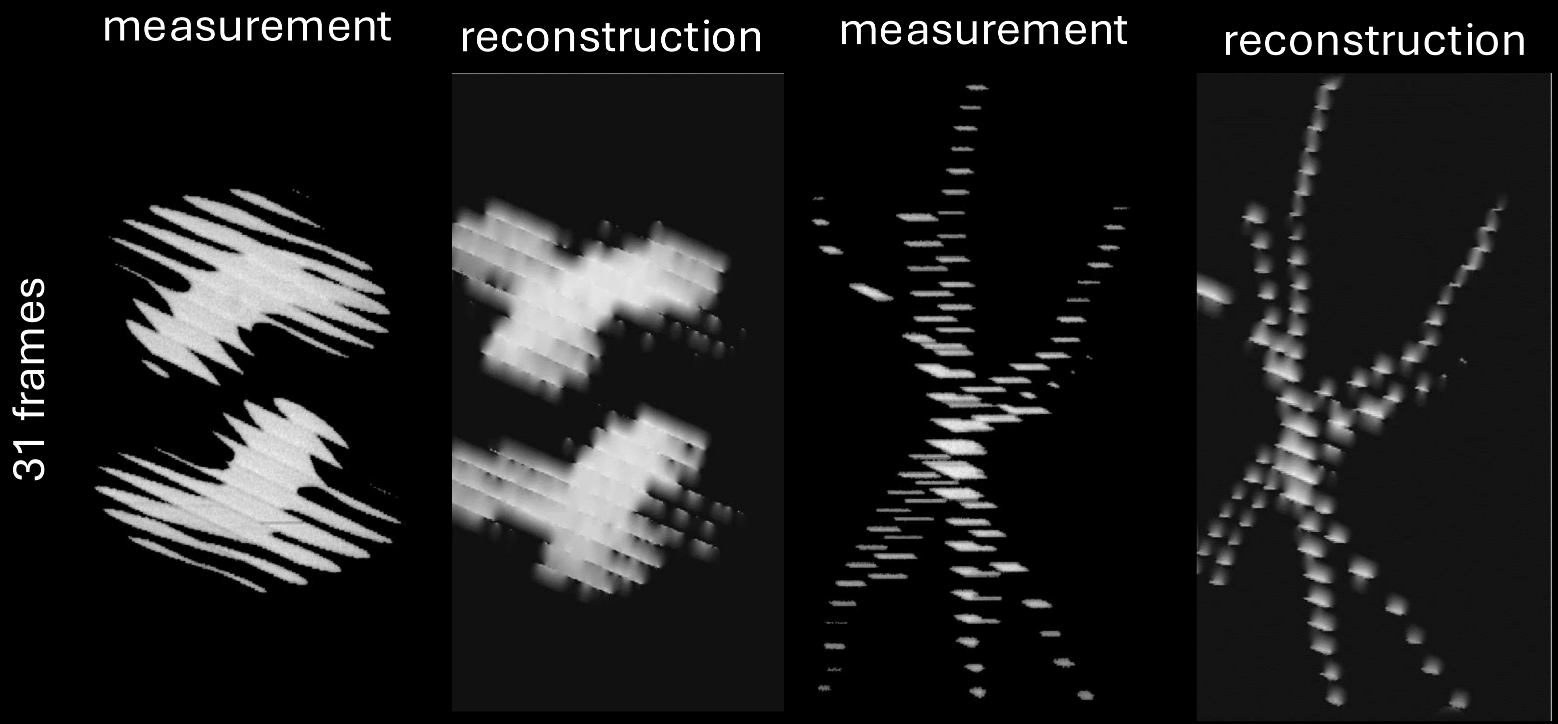
I. Amidror, “Scattered data interpolation methods for electronic imaging systems: a survey,” J. Electron. Imaging 11, 157–176 (2002).

Experimental implementation



- Architecture: compressed optical-streaking ultra-high-speed photography
- Encoding: single binary mask
- Shearing operator: Galvanometer

Real-data: Interpolation



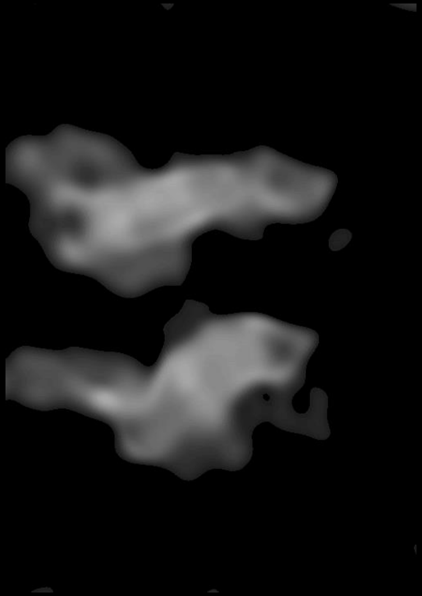
I. Amidror, "Scattered data interpolation methods for electronic imaging systems: a survey," J. Electron. Imaging 11,157–176 (2002).

Real-data with Implicit Neural Representation

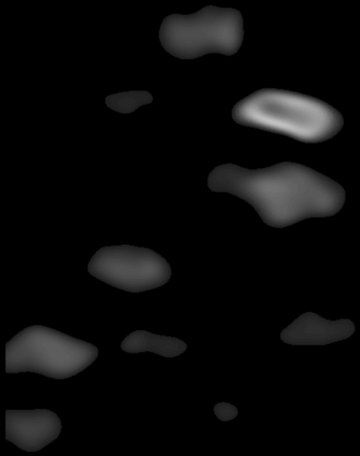
Parameters

- Initial learning rate: 0.0005
- Iteration: 1500
- Resolution: 714 x 574 x 31
- Training: a single measurement
- Training time: 1.7 s
- Testing time: 2.5 ms
- Layers: Two hidden layers

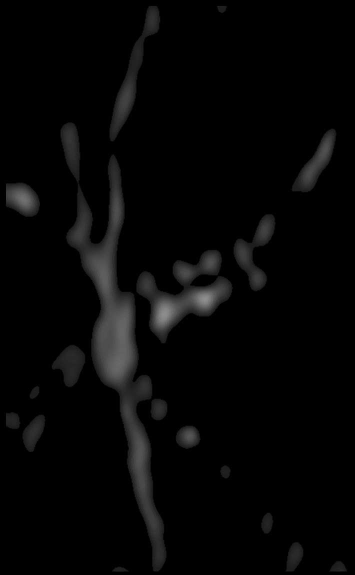
reconstruction



reconstruction



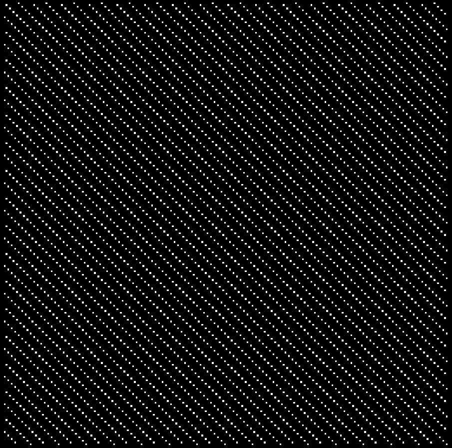
reconstruction



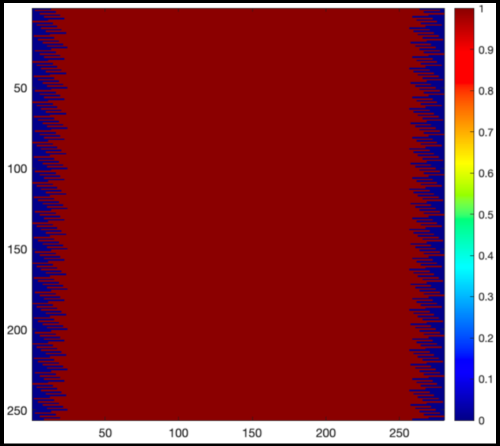
V. Sitzmann, J. Martel, A. Bergman, D. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," in Advances in Neural Information Processing Systems, vol. 33 H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, eds. (Curran Associates, Inc., 2020), pp. 7462–7473

Conclusions

- We introduced a novel binary coded aperture for compressed optical-streaking ultra-high-speed photography
- Our designed coded aperture samples uniformly the video
- We recover the underlying video using interpolation and implicit neural representation



uniform coded aperture



uniform sensing